Part A – INNER JOIN

1.

SQL

```
SELECT s.Name, d.Department_Name
FROM Student s
INNER JOIN Department d
ON s.Department_ID = d.Department_ID;
```

OUTPUT

	Name	Department_Name
▶	Aman	CSE
	Riya	CSE
	Karan	ECE
	Rahul	ME

2.

SQL

```
SELECT s.Name, c.Course_Name
FROM Student s
INNER JOIN Course c
ON s.Student_ID = c.Student_ID;
```

OUTPUT

	Name	Course_Name
▶	Aman	DBMS
	Riya	OS
	Aman	Networks

Part B – LEFT JOIN

3.

SQL

```
SELECT s.Name, d.Department_Name
FROM Student s
LEFT JOIN Department d
ON s.Department_ID = d.Department_ID;
```

OUTPUT

	Name	Department_Name
▶	Aman	CSE
	Riya	CSE
	Karan	ECE
	Sneha	NULL
	Rahul	ME

4.

SQL

```
SELECT s.Name, c.Course_Name
FROM Student s
LEFT JOIN Course c
ON s.Student_ID = c.Student_ID;
```

OUTPUT

	Name	Course_Name
▶	Aman	Networks
	Aman	DBMS
	Riya	OS
	Karan	NULL
	Sneha	NULL
	Rahul	NULL

Part C – RIGHT JOIN

5.

SQL

```
SELECT s.Name, d.Department_Name
FROM Student s
RIGHT JOIN Department d
ON s.Department_ID = d.Department_ID;
```

OUTPUT

	Name	Department_Name
▶	Riya	CSE
	Aman	CSE
	Karan	ECE
	Rahul	ME
	NULL	CE

6.

SQL

```
SELECT s.Name, c.Course_Name
FROM Student s
RIGHT JOIN Course c
ON s.Student_ID = c.Student_ID;
```

OUTPUT

	Name	Course_Name
▶	Aman	DBMS
	Riya	OS
	Aman	Networks
	NULL	AI

Part D – FULL OUTER JOIN

7.

SQL

```
SELECT s.Name, d.Department_Name
FROM Student s
LEFT JOIN Department d ON s.Department_ID = d.Department_ID
UNION
SELECT s.Name, d.Department_Name
FROM Student s
RIGHT JOIN Department d ON s.Department_ID = d.Department_ID;
```

OUTPUT

	Name	Department_Name
▶	Aman	CSE
	Riya	CSE
	Karan	ECE
	Sneha	NULL
	Rahul	ME
	NULL	CE

8.

SQL

```
SELECT s.Name, c.Course_Name
FROM Student s
LEFT JOIN Course c ON s.Student_ID = c.Student_ID
UNION
SELECT s.Name, c.Course_Name
```

```
FROM Student s
RIGHT JOIN Course c ON s.Student_ID = c.Student_ID;
```

OUTPUT

	Name	Course_Name
▶	Aman	Networks
	Aman	DBMS
	Riya	OS
	Karan	NULL
	Sneha	NULL
	Rahul	NULL
	NULL	AI

Part E – SELF JOIN

9.

SQL

```
CREATE TABLE Employee (
    Emp_ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Manager_ID INT
);

INSERT INTO Employee VALUES
(1, 'A', NULL),
(2, 'B', 1),
(3, 'C', 1),
(4, 'D', 2);

SELECT e.Name AS Employee, m.Name AS Manager
FROM Employee e
LEFT JOIN Employee m
ON e.Manager_ID = m.Emp_ID;
```

OUTPUT

	Employee	Manager
▶	A	NULL
	B	A
	C	A
	D	B

Part D – Advanced Queries

10.

SQL

```
SELECT d.Department_Name, COUNT(s.Student_ID) AS Student_Count
FROM Department d
LEFT JOIN Student s
ON d.Department_ID = s.Department_ID
GROUP BY d.Department_Name;
```

OUTPUT

	Department_Name	Student_Count
▶	CSE	2
	ECE	1
	ME	1
	CE	0

11.

SQL

```
SELECT s.Name, COUNT(c.Course_ID) AS Course_Count
FROM Student s
JOIN Course c ON s.Student_ID = c.Student_ID
GROUP BY s.Name
HAVING COUNT(c.Course_ID) > 1;
```

OUTPUT

	Name	Course_Count
▶	Aman	2

12.

SQL

```
SELECT d.Department_Name
FROM Department d
LEFT JOIN Student s
ON d.Department_ID = s.Department_ID
WHERE s.Student_ID IS NULL;
```

OUTPUT

	Department_Name
▶	CE